

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    >
    <ImageView
        android:src="@drawable/databaseicon"
        android:layout_height="wrap_content"
        android:layout_width="wrap_content"
        android:layout_gravity="center"
        android:padding="20dp"
        android:scaleType="fitCenter"
    ></ImageView>

    <TextView
        android:text="A Simple Login Example Using Database"
        android:layout_width="fill_parent"
        android:layout_height="wrap_content"
        android:textColor="#ABCEDE"
        android:scaleType="fitCenter"
    ></TextView>
</LinearLayout>
```

In this layout, we have an image and text, which we display at the launch of the application for 5 seconds. In this layout XML, the only new Android property you see from the previous article is *android:scaleType*; which is used while scaling the image. If the screen size changes, then the image will be scaled, always keeping it in the centre, so that the splash screen layout is always at the centre—even for different sizes of screen.

When the splash screen activity is completed, we fire an Intent to launch another activity called *DatabaseActivity*, which takes you to the log-in page. See examples of the splash screen and log-in page in Figures 4 and 5.

DatabaseActivity involves taking in the values, validating the input, and querying the database. The full code of this application can be found in my Github repository http://bit.ly/saket_github_loginpage. I have already explained, in the very first article in this series, about Basic Android Application Development—how to take inputs and create layouts. Therefore, I will concentrate more on the core concept rather than on the UI and basic validations.

In this activity, we set the layout to main.xml, as shown in Figure 5; we write the basic code to get inputs from the EditText boxes for usernames and passwords. We also write code for the basic validation for null strings, as we had discussed in my first article. The main validation that is required here is that of the username and password against the database. I have written a method called *validateLogin*

Simple Login Application

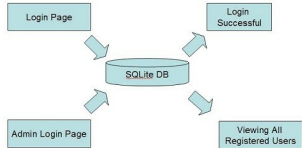


Figure 1: App design

_id	• Unique identifier of the row
username	• Holds the username
password	• Holds the password

Figure 2: Table design

that does the task.

```
Button mLogin;
Button mNewUser;
Button mShowAll;
EditText mUsername;
EditText mPassword;
DBHelper mydb = null;
```

These are the instances that we will be using in this Activity class to accomplish our objective. I created the *DBHelper* class to create and upgrade the database. We will discuss *DBHelper* in detail, as we move ahead.

```
public boolean
validateLogin(String uname,
String pass, Context context)
{
    mydb = new DBHelper(context);
    SQLiteDatabase db = mydb.getReadableDatabase();
    //SELECT
    String[] columns = {"_id"};
    //WHERE clause
    String selection = "username=? AND password=?";
    //WHERE clause arguments
    String[] selectionArgs = {uname, pass};
```

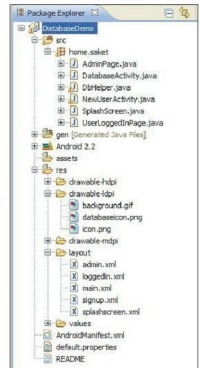


Figure 3: Folder structure